

SAHIL RATHEE

Senior Software Engineer & AI Architect

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PROFESSIONAL SUMMARY

Senior Software Engineer & AI Architect with over **4 years of experience** architecting autonomous systems, distributed cloud infrastructure, and financial engineering engines. Proven record deploying agentic AI workflows and optimized RAG architectures that **reduced operational token expenditure by 70%.** Expert in full-stack high-performance distributed systems, sub-millisecond execution engines, container orchestration, and scaling high-concurrency cross-platform ecosystems across APAC and global markets.

TECHNICAL COMPETENCIES

Languages:	TypeScript, JavaScript, Python, Go (Golang), C++, Dart, Kotlin, GraphQL, Gremlin, SQL, HTML5/CSS3
Frameworks & Web:	React, Next.js (SSR/SSG), Angular, React Native, Flutter, Cocos Creator Engine, Node.js, Express, FastAPI
Cloud & DevOps:	Kubernetes, Docker, Podman, Terraform IaC, Ansible, Jenkins CI/CD, AWS (Lambda, ECS, Neptune, OpenSearch), GCP, Cloudflare Workers
Databases & Streaming:	PostgreSQL, MySQL, MongoDB, Redis, Qdrant Vector DB, AWS Neptune, NATS JetStream, RabbitMQ, Apache Kafka
Systems & Domains:	Quantitative Research & Algorithmic Trading, High-Frequency Trading (HFT), Advanced Trading Strategies, Agentic AI Swarms, RAG Pipeline Optimization, GIS & Geospatial Indexing, Monorepo Architecture

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer — NaiBooking
Aug 2026 – Present | Full-time | Remote — Bangalore, India

- Architected full-stack salon & barber SaaS marketplace and CRM: web landing portal and admin console in **React / Next.js**, client and business mobile apps in **React Native (Expo)**.
- Published and maintained client and business apps live on **Apple App Store** and **Google Play Store**.
- Engineered Node.js backend microservices leveraging **MongoDB** and **PostgreSQL** with transaction ledgers, advisory locking, and multi-tier Redis caching to handle high-concurrency booking spikes.
- Automated AWS cloud infrastructure via **Terraform IaC**; integrated Razorpay automated subscription billing for salon CRM with an architectural roadmap for Stripe global expansion (UAE, USA, Singapore, Australia).

Tech Stack: *TypeScript, React, Next.js, React Native (Expo), Node.js, MongoDB, PostgreSQL, Redis, AWS, Terraform IaC, Docker, CI/CD*

Lead Software Engineer (Contractual) — Orealfit
May 2026 – Jul 2026 | Full-time | On-site — Sector 44, Gurugram, India

- Contractually architected and scaled an enterprise sports & wellness ecosystem featuring **real-time scorecard streaming** and low-latency interactive advertising modules.
- Transitioned engineering operations to **autonomous agentic workflows** by deploying autonomous dev and test agent pipelines, accelerating end-to-end feature delivery velocity.
- Optimized RAG pipeline efficiency and AI infrastructure through custom token usage telemetry and caching, resulting in a **70% reduction in monthly AI token expenditure**.
- Engineered a high-performance wellness e-commerce platform using monorepo architecture, streamlining cross-service asset management and deployment reliability.
- Built an edge-computed billing and GST-compliance solution for SMBs, providing real-time financial tracking with sub-50ms latency.

Tech Stack: *Flutter, Next.js, Go, C++, Node.js, PostgreSQL, Redis, Qdrant Vector DB, Supabase, Cloudflare Workers, Docker*

Career Break — Quantitative Research & Algorithmic Trading — Independent Deep-Dive
Dec 2025 – Apr 2026 | Self-Directed

- Dedicated career break focused on mastering **Quantitative Research, Advanced Trading, Algorithmic Trading, and High-Frequency Trading (HFT)** systems.
- Formulated mathematical trading models, statistical arbitrage algorithms, and tick-level backtesting simulation engines across equities and foreign exchange.
- Engineered sub-millisecond in-memory ring buffers and MetaTrader 5 execution bridges streaming real-time order-book telemetry via NATS JetStream (Zenith & Mitochondria).
- Developed invariant-based risk management safeguards including dynamic position sizing, trailing circuit breakers, and automated drawdown enforcement.

Focus & Domains: *Quantitative Research, Algorithmic Trading, High-Frequency Trading (HFT), Advanced Trading, Market Microstructure, MetaTrader 5, Python, C++, Go, NATS JetStream*

Solution Architect — Gym Wise India Private Limited — Jul 2024 – Nov 2025 | Full-time | On-site — Sector 14, Gurugram, India

- Re-architected a legacy monolithic codebase (single Kotlin Android app + PHP Laravel backend) into **3 standalone cross-platform Flutter/Dart apps** managed via build flavors: **GymWise Customer App**, **GymWise Partner App** (with biometric/attendance tracking), and **GymWise Admin App** (for manager financial, booking, and catalog management), alongside a React/Next.js web portal.
- Shipped and maintained all 3 mobile applications live on **Google Play Store** and **Apple App Store**.
- Re-engineered backend microservices from PHP Laravel to **Go (Golang)** backed by a Redis caching buffer and **Qdrant Vector DB** for member search and similarity matching.
- Built multi-cloud infrastructure: **AWS** for compute and container orchestration managed declaratively via **Terraform IaC and Ansible**; **GCP** for Maps APIs and OAuth services.
- Implemented gym marketplace business logic: zero upfront cost to gym owners, automated **30% commission** collection on customer service bookings, and algorithmic dynamic pricing based on user engagement and demand.

Tech Stack: *Flutter, Dart, Go (Golang), AWS, GCP, Redis, Qdrant Vector DB, PostgreSQL, Terraform IaC, Ansible, Docker Swarm, Play Console, App Store Connect*

Career Break — AI Research & Agentic Systems — Independent Deep-Dive — Apr 2024 – Jun 2024 | Self-Directed

- Dedicated career break between BigHit and GymWise to conduct comprehensive research into **Autonomous AI Agents and Multi-Agent Workflows**.
- Researched and built AST context compression algorithms, dynamic prompt caching, and isolated Docker execution environments for autonomous coding agents (Meeseeks framework).
- Designed hybrid agentic RAG retrieval architectures, hierarchical vector indexing strategies, and automated decision verification loops.

Focus & Domains: *Autonomous AI Agents, Multi-Agent Swarms, AST Code Parsing, Vector Embeddings, LLM Orchestration, Python, Docker*

Full Stack Engineer — Bighit — Nov 2022 – Mar 2024 | Full-time | On-site — Andheri West, Mumbai, India

- Developed real-time **Socket.io Node.js server** powering dynamic chat for **2,000 Concurrent Active Users (CCU)** (~400 events/sec peak throughput).
- Built real-time score updates and live telemetry engine for **1,000 CCU** live-streamed football matches (U-14 tournaments, BigHit Sportz), featuring a dedicated third-umpire web console with access control and abstracted scoring workflows (akin to CricHeroes).
- Migrated legacy relational SQL database to **Amazon Neptune graph DB (Gremlin)**, optimizing athlete-to-scout social graph traversal and talent discovery queries.
- Modernized mobile and backend infrastructure (React Native & AWS IaC), achieving an **~80% reduction in app bundle size** and an **~60% increase in server efficiency**.
- Integrated and managed AWS cloud infrastructure with **Amazon Neptune** graph DB and **OpenSearch** clusters, reducing API response latency by **80%**.
- Led and mentored a team of 6 engineers: ran technical roadmaps, sprint planning, code reviews, and developer onboarding.

Tech Stack: *TypeScript, React Native, Node.js, Socket.io, GraphQL, AWS Neptune (Graph DB), OpenSearch, Docker, Kubernetes, AWS IaC, CI/CD*

Full Stack Developer — Gamix Labs — Mar 2022 – Oct 2022 | Full-time | On-site — Noida, India

- Engineered real-time mathematical simulation engines for Vegas slot machines (HTML5/Facebook Instant Games on Cocos Creator Engine) with Node.js and RabbitMQ event streaming backends.
- Formulated and verified Return-to-Player (RTP) and win frequency probability algorithms via Monte Carlo simulations (**10,000,000+ spins at 99.9% confidence interval** across win bands: 65.2% zero-win, 15.4% <1x, 12.1% 1-5x, 5.2% 5-20x, 1.8% 20-100x, 0.3% 100x+) ensuring statistical house edge while maintaining user engagement.
- Architected multi-database backends across client games utilizing both **SQL** and **NoSQL** engines (**MongoDB** and **Firebase Realtime Database / Firestore**) tailored to game mechanics.
- Collaborated with tier-1 international gaming clients (OneFC, Bombay Play) to deliver responsive PWA online gaming platforms and HTML5 titles with Redis and MongoDB caching.
- Developed a cross-platform biometric facial and voice recognition verification app for online interview security.

Tech Stack: *TypeScript, Cocos Creator Engine, Node.js, RabbitMQ, MongoDB, Firebase, SQL, Redis, Angular, Flutter, Java, Kotlin, GCP*

KEY ENGINEERING & OPEN-SOURCE INITIATIVES

Zenith & Mitochondria — Autonomous Forensic Trading Workstation — Infortts Open-Source Swarm

- Architected an autonomous quantitative trading engine integrated with MetaTrader 5, NLP market sentiment analysis, and vectorized trade analytics.
- Engineered sub-millisecond in-memory ring buffers and custom backtesting simulation engines streaming live telemetry via NATS JetStream to a Flutter workstation.

Tech Stack: *Python, C++, Go, NATS JetStream, PostgreSQL, Qdrant Vector DB, WebSockets, Flutter*

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science — DDUC, University of Delhi — 2018 – 2021 | New Delhi, India

INTERESTS & SPECIALIZATIONS

High-Frequency Trading (HFT) Architecture • Algorithmic Strategy & Quantitative Modeling • Distributed Multi-Agent Systems • Competitive Gaming (CS:GO, Chess)